

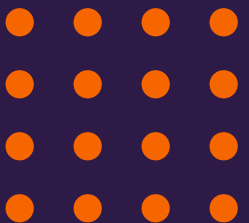


Outlier Detection & Analysis

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Outlier Analysis - Outline

Introduction / Motivation / Definition

Statistical-based Detection

- Distribution-based, depth-based

Deviation-based Detection

- Sequential exception, OLAP data cube

Distance-based Detection

- Index-based, nested-loop, cell-based, local-outliers

Questions



Introduction

Traditional Data Mining Categories

- Majority of Objects
 - Dependency detection
 - Class identification
 - Class description
- Exceptions
 - Exception/outlier detection



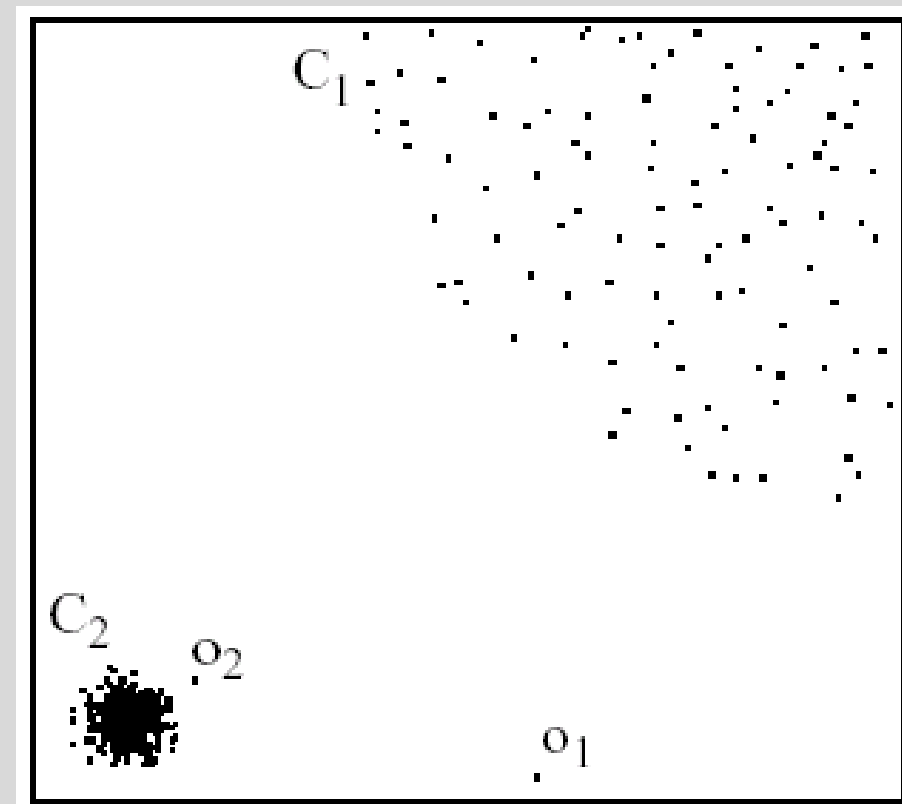
Motivation for Outlier Analysis

- Fraud Detection
(Credit card, telecommunications, criminal activity in e-Commerce)
- Customized Marketing (high/low income buying habits)
- Medical Treatments (unusual responses to various drugs)
- Analysis of performance statistics (professional athletes)
- Weather Prediction
- Financial Applications (loan approval, stock tracking)

“One persons noise could be another person’s signal.”

What is an outlier?

- Observations inconsistent with rest of the dataset – Global Outlier
- Special outliers – Local Outlier
 - Observations inconsistent with their neighborhoods
 - A local instability or discontinuity





Causes of Outliers

- Poor data quality / contamination
- Low quality measurements
- Malfunctioning equipment
- Manual error
- Correct but exceptional data
- Many more!



Outlier Detection Approaches

Objective:

- Define what data can be considered as inconsistent in a given data set
 - Statistical-Based Outlier Detection
 - Deviation-Based Outlier Detection
 - Distance-Based Outlier Detection
- Find an efficient method to mine the outliers



Why A Special Technique to Identify Outliers?

Why not just modify clustering or other algorithms to detect outliers?

- Performance considerations
- Subjective to the clustering algorithm and clustering parameters
- Only certain attributes may have outlier properties, no need to disqualify the entire tuple
- Contamination may occur by “column”, not by row



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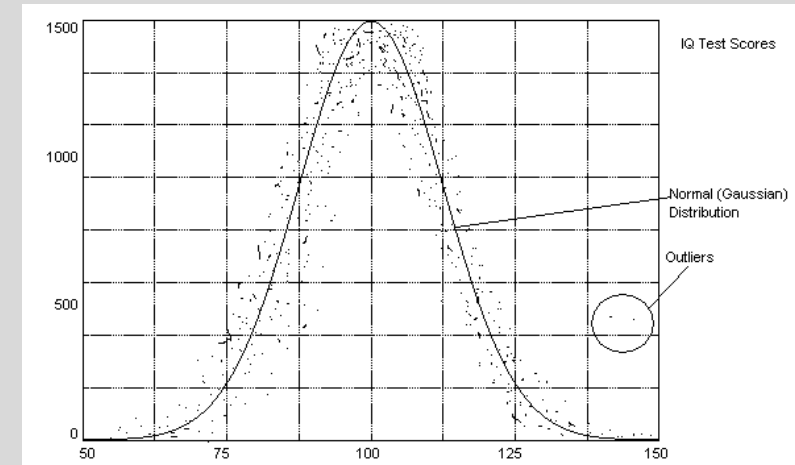
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Statistical-Based Outlier Detection (Distribution-based)

- Assumptions:
 - Knowledge of data (distribution, mean, variance)
- Statistical discordancy test
 - Data is assumed to be part of a working hypothesis (working hypothesis)
 - Each data object in the dataset is compared to the working hypothesis and is either accepted in the working hypothesis or rejected as discordant into an alternative hypothesis (outliers)



Working Hypothesis: $H: o_i \in F$, where $i = 1, 2, \dots, n$.

Discordancy Test: is o_i in F within standard deviation = 15

Alternative Hypothesis:

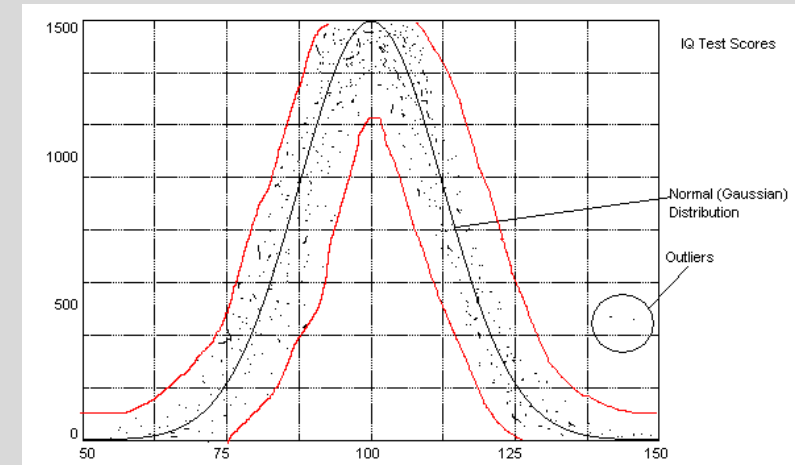
-Inherent Distribution: $\bar{H}: o_i \in G$, where $i = 1, 2, \dots, n$.

-Mixture Distribution: $\bar{H}: o_i \in (1 - \lambda)F + \lambda G$, where $i = 1, 2, \dots, n$.

-Slippage Distribution: $\bar{H}: o_i \in (1 - \lambda)F + \lambda F'$, where $i = 1, 2, \dots, n$.

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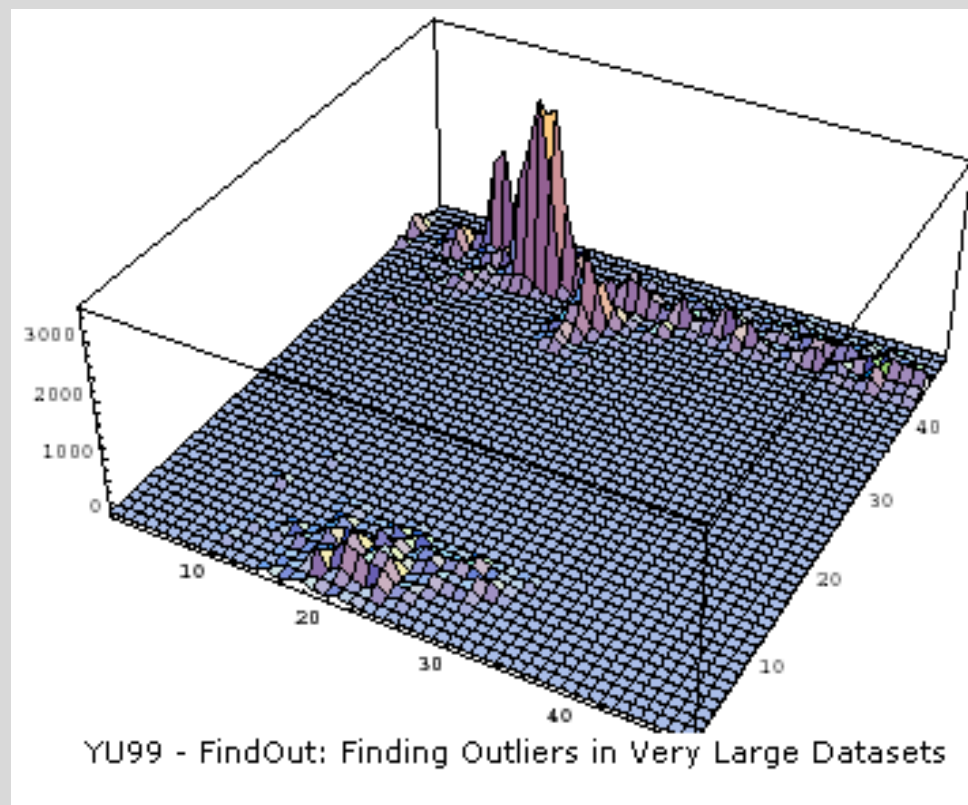
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Statistical-Based Outlier Detection (Depth-based)

- Data is organized into layers according to some definition of depth
- Shallow layers are more likely to contain outliers than deep layers
- Can efficiently handle computation for $k < 4$



Statistical-Based Outlier Detection

- Strengths
 - Most outlier research has been done in this area, many data distributions are known
- Weakness
 - Almost all statistical models are univariate (only handle one attribute) and those that are multivariate only efficiently handle $k < 4$
 - All models assume the distribution is known – this is not always the case
 - Outlier detection is completely subjective to the distribution used



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Deviation-Based Outlier Detection

Simulate a mechanism familiar to human being: after seeing a series of similar data, an element disturbing the series is considered an exception

Sequential Exception Techniques

OLAP Data Cube Techniques

Sequential Exception

- Select subsets of data I_j ($j = 1, 2, \dots, n$) from the dataset I
- Compare the dissimilarity of I and $(I - I_j)$ with $\sum_{i=1}^N (x_i - \bar{x})^2$
- Find out the minimum subset I_j that reduce the dissimilarity the most
- Smoothing factor $SF(I_j) := C(I - I_j) * (D(I) - D(I - I_j))$
 - D is a dissimilarity function
 - C is a cardinality function, for example, the number of elements in the dataset $\frac{1}{N} X$

Example

Let the data set I be the set of integer values $\{1, 4, 4, 4\}$

I_j	$I - I_j$	$C(I - I_j)$	$D(I - I_j)$	$SF(I_j)$
$\{\}$	$\{1, 4, 4, 4\}$	4	1.69	0.00
$\{4\}$	$\{1, 4, 4\}$	3	2.00	-0.93
$\{4, 4\}$	$\{1, 4\}$	2	2.25	-1.12
$\{4, 4, 4\}$	$\{1\}$	1	0.00	1.69
$\{1\}$	$\{4, 4, 4\}$	3	0.00	5.07
$\{1, 4\}$	$\{4, 4\}$	2	0.00	3.38
$\{1, 4, 4\}$	$\{4\}$	1	0.00	1.69

Note, when $I_j = \{\}$, $D(I) = D(I - I_j) = 1.69$, $SF(I_j) = 0$

When $I_j = \{1\}$, $SF(I_j)$ has the maximum value, so $\{1\}$ is the outlier set



OLAP Data Cube Technique

- Deviation detection process is overlapped with cube computation
- Precomputed measures indicating data exceptions are needed
- A cell value is considered an exception if it is significantly different from the expected value, based on a statistical model
- Use visual cues such as background color to reflect the degree of exception



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Distance-Based Outlier Detection

Distance-based: An object O in a dataset T is a $DB(p,D)$ outlier if at least fraction p of the objects in T are $\geq$ distance D from O

A point O in a dataset is an outlier with respect to parameters k and d if no more than k points in the dataset are at a distance of d or less from O .

Relative measurement: Let $D^k(O)$ denote the distance of the k^{th} nearest neighbor of O . It is a measure of how much of an outlier point O is.

Index-Based Algorithm [KN98]

Indexing Structures such as R-tree (R+-tree), K-D (K-D-B) tree are built for the multi-dimensional database

The index is used to search for neighbors of each object O within radius D around that object.

Once K ($K = N(1-p)$) neighbors of object O are found, O is not an outlier.

Worst-case computation complexity is $O(K * n^2)$, K is the dimensionality and n is the number of objects in the dataset.

Pros: scale well with K

Cons: the index construction process may cost much time



Nested-Loop Algorithm [KN98]

Divides the buffer space into two halves (first and second arrays)

Break data into blocks and then feed two blocks into the arrays.

Directly computes the distance between each pair of objects, inside the array or between arrays

Decide the outlier.

Here comes an example:...

Same computational complexity as the index-based algorithm

Pros: Avoid index structure construction

Try to minimize the I/Os



Example – stage 1

Buffer

A
B

DB

A	B
C	D

Starting Point of Stage 1

A
D

A	B
C	D

End Point of Stage 1

A is the target block on stage 1

Load A into the first array (1R)

Load B into the second array (1R)

Load C into the second array (1R)

Load D into the second array (1R)

Total: 4 Reads



Example – stage 2

Buffer

A
D

DB

A	B
C	D

Starting Point of Stage 2

C
D

A	B
C	D

End Point of Stage 2

D is the target block on stage 2

D is already in the buffer (no R)

A is already in the buffer (no R)

Load B into the first array (1R)

Load C into the first array (1R)

Total: 2 Reads



Example – stage 3

Buffer

C
D

DB

A	B
C	D

Starting Point of Stage 3

C
B

A	B
C	D

End Point of Stage 3

C is the target block on stage 3

C is already in the buffer (no R)

D is already in the buffer (no R)

Load A into the second array (1R)

Load B into the second array (1R)

Total: 2 Reads



Example – stage 4

Buffer

C
B

DB

A	B
C	D

Starting Point of Stage 4

D
B

A	B
C	D

End Point of Stage 4

B is the target block on stage 4

B is already in the buffer (no R)

C is already in the buffer (no R)

Load A into the first array (1R)

Load D into the first array (1R)

Total: 2 Reads

Every block is $\frac{1}{4}$ of the DB. From stage 1-4, a grand total of 10 blocks are read, amounting to $10/4$ passes over the entire dataset.

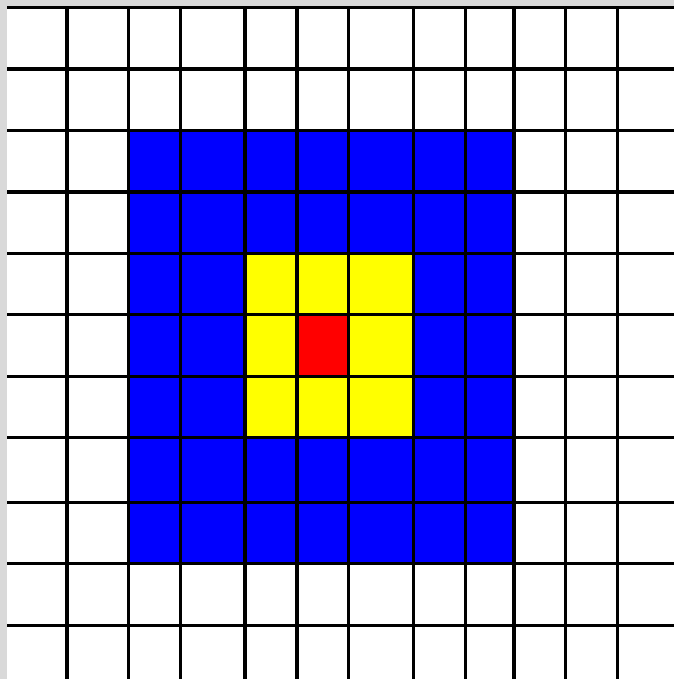


Cell-Based Algorithm [KN98]

- Divide the dataset into cells with length $l = \frac{D}{2\sqrt{k}}$
 - K is the dimensionality, D is the distance
- Define Layer-1 neighbors – all the intermediate neighbor cells.
The maximum distance between a cell and its neighbor cells is D
- Define Layer-2 neighbors – the cells within 3 cell of a certain cell.
The minimum distance between a cell and the cells outside of Layer-2 neighbors is D
- Criteria
 - Search a cell internally. If there are M objects inside, all the objects in this cell are not outlier
 - Search its layer-1 neighbors. If there are M objects inside a cell and its layer-1 neighbors, all the objects in this cell are not outlier
 - Search its layer-2 neighbors. If there are less than M objects inside a cell, its layer-1 neighbor cells, and its layer-2 neighbor cells, all the objects in this cell are outlier
 - Otherwise, the objects in this cell could be outlier, and then need to calculate the distance between the objects in this cell and the objects in the cells in the layer-2 neighbor cells to see whether the total points within D distance is more than M or not.
- An example:



Example



Red – A certain cell

Yellow – Layer-1 Neighbor Cells

Blue – Layer-2 Neighbor Cells

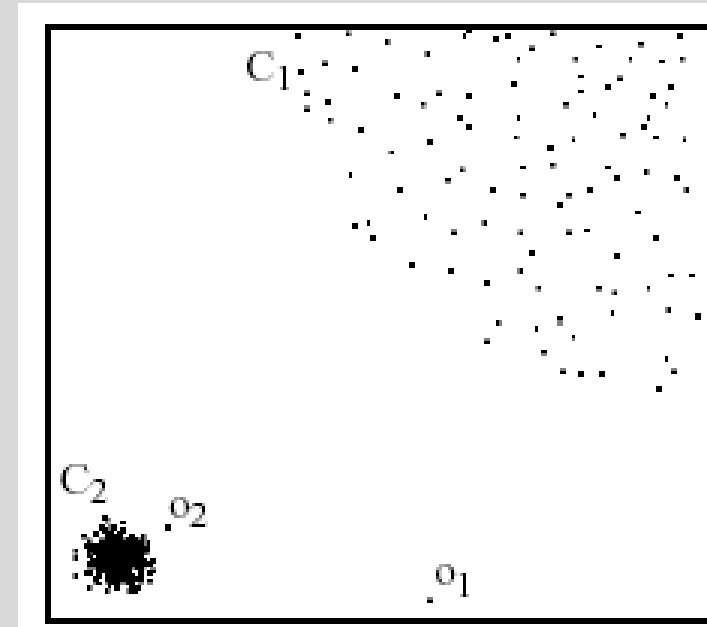
Notes:

The maximum distance between a point in the red cell and a point in its layer-1 neighbor cells is D

The minimum distance between a point in the red cell and a point outside its layer-2 neighbor cells is D

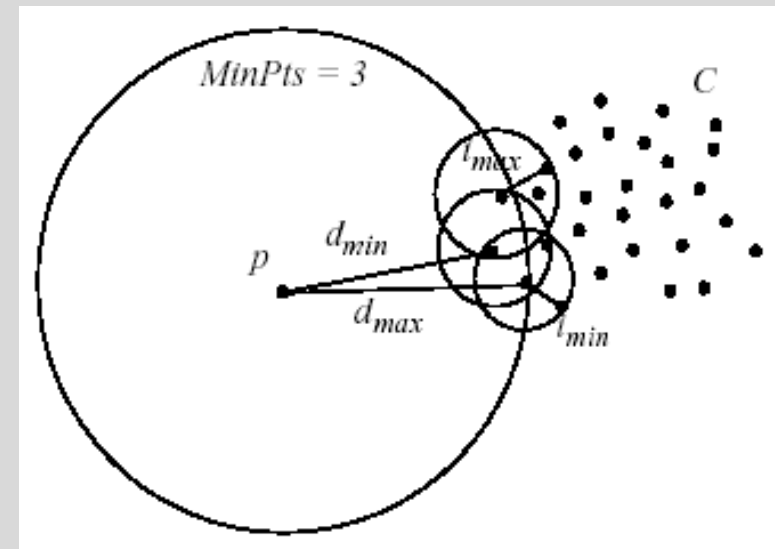
Distance-Based Outlier Detection (Local Outliers)

- Some outliers can be defined as global outliers, some can be defined as local outliers to a given cluster
- O_2 would not normally be considered an outlier with regular distance-based outlier detection, since it looks at the global picture



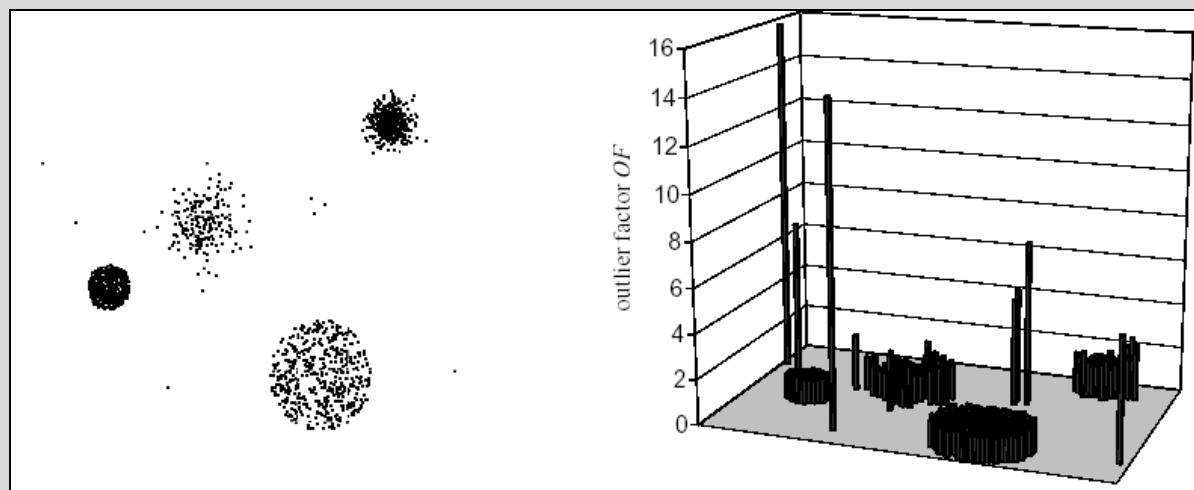
Distance-Based Outlier Detection (Local Outliers)

- Each data object is assigned a *local outlier factor (LOF)*
- Objects which are closer to dense clusters receive a higher LOF
- LOF varies according to the parameter MinPts





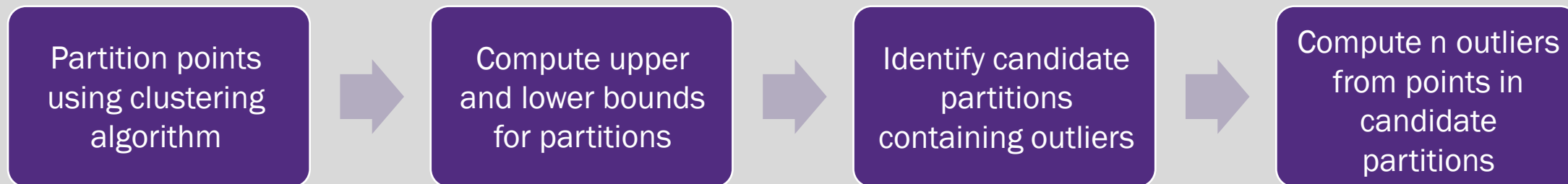
Distance-Based Outlier Detection (Local Outliers)





Distance-Based Outlier Detection (Partition-based)

- Partition-based detection
 - Use BIRCH clustering to identify clusters/partitions of non-outliers
 - Prune partitions that do not contain outliers
 - Use Index/Nested Loop algorithms on the remaining data points
 - Since many data point are removed during pruning, the efficiency is increased significantly.





Questions?